

Deepanjay Nandal

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SUMMARY

Senior Software Engineer with 3+ years building production systems, from CVE validation infrastructure for Google's Android Security team to LLM inference caching and event-driven pipelines processing 100k+ daily events. M.S. from Arizona State University. Combines low-level systems and security depth with hands-on AI infrastructure work across semantic caching, hybrid retrieval, and Kafka/Redis pipelines. Focused on low-latency, fault-tolerant distributed systems.

EXPERIENCE

Discover Excellence LLC

California, USA

Software Engineer (Contract)

Jan 2026 – Jun 2026

- Built a high-throughput Kafka-based visitor intelligence pipeline for an AI-powered eCommerce analytics platform, processing 100k+ daily behavioral events at sub-40ms p95 latency to drive discount outreach via AWS SES.
- Designed a Redis-backed collaborative filtering layer for real-time in-memory product affinity scoring.
- Served personalized product feeds at the edge via Cloudflare, achieving 92% cache hit rate and sub-30ms p95 latency for multi-tenant deployments.
- Implemented a fault-tolerant BullMQ background pipeline for async job processing, handling 50k+ daily jobs with guaranteed job delivery through persistent state, retries, exponential backoff, and idempotency keys.

Ittiam Systems Pvt Ltd

Bengaluru, India

Senior Software Engineer

Mar 2022 – Jun 2024

- Authored 50+ CTS/STS validation tests merged into the Android Open Source Project (AOSP), each proving a CVE patch for Google's Android Security team, working directly with Google engineers.
- Developed a Ghidra-based binary analysis approach for the Android Bluetooth stack that Google had marked infeasible, which the team adopted to validate 40+ Bluetooth CVEs.
- Engineered a Frida instrumentation framework in Python and Java to hook private Android methods at runtime, enabling the team to reproduce 30+ CVEs that standard testing could not reach.
- Built a GCP Python pipeline ingesting ClusterFuzz crash outputs across Dockerized environments, auto-classifying reports by CVSS severity into a PostgreSQL-backed schema, cutting manual triage overhead by 35%.
- Automated webhook-driven Gerrit CI/CD workflows with Change-Id-keyed idempotency, processing 30–40 CVE submissions per month without duplicate change creation.

PROJECTS

BitMod (AI Inference Cache) | *Python, FastAPI, Vector Embeddings, Multi-tenant, Docker*

Jan – Jun 2026

- Architected a 9-layer LLM cache proxy with SHA-256 exact lookup, semantic reuse, and Bayesian evidence scoring, achieving up to 94% hit rate and 176x speedup at 71ms cached latency.
- Established multi-tenant namespace isolation, intent-based cache routing with full LLM bypass for deterministic queries, and a knowledge ingestion pipeline that eliminates cold start by pre-seeding 50k+ cached Q&A pairs.

CaseTally (Legal Search Platform) | *FastAPI, pgvector, RAG, BM25, LLM APIs*

Jun – Dec 2025

- Constructed a hybrid legal search system over 33k+ U.S. Code chunks using BM25 and pgvector, with LLM query rewriting improving MRR by 21% and weighted score fusion across both retrieval paths.
- Built an idempotent ingestion pipeline across 54 U.S. Code titles with SHA256-based change detection and a FastAPI SSE streaming backend, with raw retrieval benchmarked at p50 6ms and max 22ms.

WeVibe | *Backend Engineer (Contract), Appy.io | Node.js, PostgreSQL, Socket.IO, Redis*

Jan – Apr 2026

- Created a real-time matchmaking backend using Socket.IO and Redis pub/sub for horizontally scalable event delivery, PostgreSQL advisory locks preventing race conditions, and PostGIS geo-filtering on Google Cloud Run.
- Integrated Gemini bio generation with a custom prompt-injection guard and rate limiting (5 bios/day, 60s cooldown), Firebase JWT auth, and Apple Sign-In with revocation, validated across 17 Jest integration suites.

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, SQL

Backend Technologies: Distributed Systems, Microservices, Event-Driven Architecture, Node.js, Multi-tenancy, REST APIs, WebSockets, Message Queues, BullMQ, Rate Limiting, Socket.IO

Data & Infrastructure: Apache Kafka, PostgreSQL, PostGIS, Redis, pgvector, Distributed Caching

Security & Systems: Android Open Source Project (AOSP), Binary Analysis, Ghidra, Frida, ClusterFuzz

Cloud & DevOps: AWS (EC2, Lambda, SQS, IoT Greengrass, SES), GCP, Firebase, Cloudflare, Docker, CI/CD

AI Infrastructure: Retrieval-Augmented Generation (RAG), Vector Search, Large Language Model (LLM) APIs, Semantic Caching, Embeddings, Collaborative Filtering, Retrieval Evaluation

EDUCATION

Arizona State University

Master of Science (M.S.) in Computer Software Engineering

Tempe, Arizona, USA

Aug 2024 – May 2026

Vellore Institute of Technology

Bachelor of Technology (B.Tech) in Information Technology

Tamil Nadu, India

Jul 2017 – Jul 2021